Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Beat Optical, and Does It Hold Across ViT and CNN?

Anonymous WACV Applications Track submission

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Abstract

Detecting "dark" vessels—ships not broadcasting their Automatic Identification System (AIS) signal—in spaceborne synthetic aperture radar (SAR) is an operationally important, label-scarce problem. Pretrained backbones are a natural remedy, but it is unclear which pretraining recipe transfers best to this regime, and whether the answer is the same for Transformer and convolutional backbones. We propose a controlled label-efficiency study on the xView3-SAR benchmark [8] that holds a single point-native detector fixed and varies only the downloaded backbone initialization. The study spans two architecture-matched tracks—a ViT-B/16 track (∼86M parameters) and a size-matched ConvNeXt-V2-Base track (∼89M)—each with four arms: random initialization, optical-domain pretraining, SAR-domain pretraining, and supervised out-of-domain SAR pretraining. The headline deliverable is a two-track label-efficiency curve that (i) quantifies how much labeled data each pretraining recipe saves, (ii) tests whether SAR-domain pretraining wins in the scarce-label regime, and (iii) tests whether that ordering is architecture-general. Because we compare released checkpoints rather than pretraining the foundation models ourselves, differences are attributable to the pretraining source rather than to our own optimization choices.

1. Introduction

Detecting ships that are not broadcasting their AIS signature—"dark vessels"—in spaceborne SAR is a critical maritime-monitoring problem, and one where labeled data is scarce: the xView3-SAR benchmark [8] provides Sentinel-1 imagery, but its human-verified dark-vessel labels are few, so supervised-only training tends to overfit.

Pretrained backbones can help, but it is unclear which pretraining recipe transfers best to dark-vessel detection: self-supervised pretraining on generic optical satellite imagery (fMoW-RGB [2], via Sat-DINO [9]), self-supervised pretraining on large-scale SAR (SAR-1M, via SARMAE [7]), or supervised detection pretraining on labeled but out-of-domain SAR (LS-SSDD-v1.0 [11]). It is equally unclear whether the best recipe is the same for a Transformer (ViT [4]) and a convolutional (ConvNeXt-V2 [10]) backbone.

We compare publicly released checkpoints rather than pretraining our own, for a methodological reason rather than a compute one: our question is which pretraining dataset/recipe transfers best, so each arm must faithfully represent its recipe as published. Repretraining these models ourselves would fold our own optimization choices into the comparison and confound the exact variable we want to isolate. Fixing the detector, the fine-tuning schedule, and the evaluation then lets any performance gap be attributed to the pretraining source alone. This situates the work in the pretraining-versus-self-training literature [13], specialized to dark-vessel, scarce-label SAR detection.

Research questions. With the detector, fine-tuning schedule, and input representation held fixed, we ask:

- Q1 Relative to random initialization, how much labeled xView3 data does each pretraining recipe save to reach a target dark-vessel detection F1?
- Q2 In the scarce-label regime, does SAR-domain pretraining transfer better than optical-domain pretraining?
- Q3 Is the answer architecture-general—does the same ordering hold for both a ViT and a size-matched CNN backbone?

Hypothesis. Within each architecture, SAR-domain pretraining yields the steepest label-efficiency gains and the largest dark-vessel-recall advantage at low label budgets (ordering, per track, roughly random < optical $\lesssim$ supervised < SAR, converging as labels grow), and this ordering is architecture-general.

Contributions. (1) A controlled, metric-matched protocol that isolates the pretraining recipe from architecture and optimizer choices; (2) the first two-track (ViT and CNN) label-efficiency comparison of optical- versus SAR-domain pretraining for dark-vessel detection; and (3) a point-native detector and evaluation aligned to the xView3 metric, released with continuous integration checks that enforce the study's comparability invariants.

2. Approach

2.1. Study design: two tracks, eight arms

The study has two architecture-matched tracks: a ViT track (ViT-B/16, ~86M parameters) and a CNN track (ConvNeXt-V2-Base, ~89M, deliberately size-matched). Within each track, all four arms share the same backbone, detection head, optimizer, schedule, and seeds; only the downloaded initialization differs. Initialization is the only variable within a track, and architecture the only variable across matched roles, so comparing arm k to arm $k{+}4$ isolates the architecture family with the pretraining role held fixed (Table 1).

2.2. Three pretraining regimes (data types)

The arms differ in the kind of data their backbone was pretrained on, and these differ in modality, labels, and domain:

- Optical (SatDINO, DINO self-distillation [1], on fMoW-RGB; BigEarthNet-S2): multispectral or RGB reflectance imagery of the land surface. It is visually rich and human-interpretable but is a different sensing modality from SAR—no speckle, different geometry and scattering physics.
- SAR-domain, self-supervised (SARMAE, a masked autoencoder [5], on SAR-1M; BigEarthNet-S1): Sentinel-1 amplitude imagery—the same modality as the target, with the same speckle statistics and geometry, but pretrained self-supervised (no detection labels) on a different scene distribution.
- Labeled out-of-domain SAR (LS-SSDD): Sentinel-1 imagery with ship-detection labels, i.e. the same modality and a related supervised task, but a different acquisition/scene distribution than xView3.

Contrasting these regimes is precisely what separates "same pixels, different task" from "different pixels, same task," and is the axis Q2 measures.

2.3. Detector

xView3 labels are points, and its official metric matches predictions to ground truth by distance, not box IoU. We therefore use a point-native detector: a heatmap head in the style of CenterNet / Objects as Points [12], where each vessel is a Gaussian blob on a stride-4 response map, decoded by peak-finding with distance-based non-maximum suppression. This mirrors the metric exactly and avoids synthesizing bounding boxes the data never had. A single backbone-to-head adapter maps either backbone's feature map to the shared head, so the head, loss, optimizer, and schedule are identical across both tracks.

2.4. Input representation

Every arm receives the same fixed 3-channel input, $[\mathrm{VH}, \mathrm{VV}, \mathrm{VH}{-}\mathrm{VV}]$ in dB, so the input is physically identical across arms and channel handling is never a confound. VH and VV are distinct polarimetric measurements and their difference is a physically meaningful third channel. Each backbone's stem adapts to this input through the standard timm in_chans path.

2.5. Fine-tuning and comparability guarantees

All arms are fine-tuned end-to-end through one PyTorch Lightning code path with the same optimizer (AdamW), layer-wise learning-rate decay, augmentation, and seeds. We train each arm until validation performance stops improving—early stopping on the held-out dev split, halting after a fixed patience once dev F1 plateaus—rather than for a fixed number of epochs, so every arm is compared at its own best operating point rather than at an arbitrary epoch. The decision threshold is selected on the dev split and frozen before touching the test or final-evaluation splits.

Because the validity of the comparison rests on the arms being genuinely comparable, we enforce it mechanically rather than by convention. Continuous integration runs guard tests on every change: a backbone-parity test asserts that all four arms within a track instantiate the identical parameter count (ViT-B/16 for arms 1–4, ConvNeXt-V2-Base for arms 5–8) and that both tracks' head adapters emit the same output geometry; a checkpoint-load test asserts each downloaded backbone loads with its pretrained weights (not silently random); and a scorer-immutability test freezes the evaluation code. These make "the only thing that changed is the initialization" a checked invariant, not a claim.

2.6. Pipeline

Every arm follows the same three stages (Figure 1): a pretrained backbone is downloaded, fine-tuned on

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Arm	Track	Backbone	Initialization	Params	On curve?	Role
1	ViT	ViT-B/16	random init	$\sim$86M	yes	ViT floor
2	ViT	ViT-B/16	SatDINO (fMoW-RGB, DINO)	$\sim$86M	yes	optical-domain transfer
3	ViT	ViT-B/16	SARMAE (SAR-1M, MAE)	$\sim$86M	yes	SAR-domain transfer
4	ViT	ViT-B/16	LS-SSDD supervised (we train)	$\sim$86M	yes	labeled out-of-domain SAR
5	CNN	ConvNeXt-V2-B	random init	$\sim$89M	yes	CNN floor
6	CNN	ConvNeXt-V2-B	BigEarthNet-S2 (optical RS)	$\sim$89M	yes	optical-domain transfer
7	CNN	ConvNeXt-V2-B	BigEarthNet-S1 (SAR RS)	$\sim$89M	yes	SAR-domain transfer
8	CNN	ConvNeXt-V2-B	LS-SSDD supervised (we train)	$\sim$89M	yes	labeled out-of-domain SAR
R1	ref	ConvNeXt-V2-B	ImageNet (contingent)	$\sim$89M	no	generic-transfer reference
R2	ref	YOLO26	COCO-pretrained	external	no	detector reference
R3	ref	LocateAnything-3B	zero-shot	$\sim$3B	no	VLM reference

Table 1. The eight controlled arms (two size-matched tracks) plus three external references. Params are backbone parameter counts; the ViT and CNN backbones are deliberately size-matched ($\sim$86M vs. $\sim$89M) so the cross-track comparison is fair. Optical anchors are SatDINO [9] (ViT) and BigEarthNet-S2 [3] (CNN); SAR anchors are SARMAE [7] (ViT) and BigEarthNet-S1 [3] (CNN). References R1–R3 are reported separately and are not on the controlled curves.

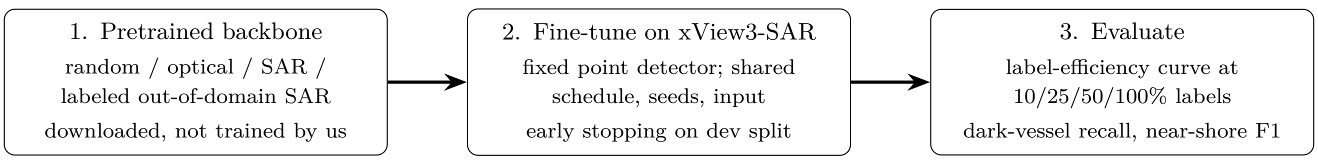

Figure 1. The three stages of the study, repeated identically for every arm. Only stage 1 (the downloaded backbone) varies within a track.

xView3-SAR through the fixed detector, and evaluated on the label-efficiency curve. Only stage 1 changes across arms within a track.

3. Datasets

xView3-SAR. We use xView3-SAR [8] via the SARFish mirror (Sentinel-1 GRD, dual-polarization VH/VV). We fine-tune on a subset of xView3 training scenes—75–150 scenes, chipped at 800×800 with overlap, yielding on the order of 10^4 chips—and sub-sample at the scene level to 10/25/50/100% of labels, with nested fractions so the curves are monotone in data. The 75–150 scenes we fine-tune on are split 75/15/10 (train/dev/test) with no scene in more than one split; separately and disjointly, the $\sim$50 human-verified validation scenes are held out as a final-evaluation set and touched exactly once. Detection positives are vessels of HIGH/MEDIUM confidence; a vessel is treated as "dark" when its source is manual (no AIS correlate).

LS-SSDD-v1.0. The supervised out-of-domain SAR arms (Arms 4 and 8) pretrain on LS-SSDD-v1.0 [11]: native Sentinel-1 imagery cut into $\sim$9,000 labeled 800×800 sub-images. LS-SSDD is used only as a pretraining source and is never mixed into the xView3 splits; its boxes are reduced to centroids so it can drive the same point-native head.

4. Evaluation Plan

The headline deliverable is a two-track label-efficiency curve (Figure 2): detection F1 versus 10/25/50/100% of real labels, eight lines (solid ViT, dashed CNN; one color per pretraining role). Within a track, the low-label gap over the random-init floor quantifies pretraining value and the SAR-versus-optical ordering (Q1, Q2); across tracks, matched-role gaps test architecture-generality (Q3). We additionally report dark-vessel recall and near-shore F1—the operationally hardest slices, where any SAR advantage should concentrate—and situate the eight arms against the external references (YOLO26 [6]; a zero-shot VLM; a contingent ImageNet-ConvNeXt).

5. Timeline, Compute, and Feasibility

Two contributors work largely independent lanes (data/scorer; detector/eval) with no hard dependency, since all foundation-model backbones are downloaded and there is no pretraining handoff to block on. Compute is one 8×V100 (32 GB) node plus a development GPU. Because we pretrain no foundation model, the total is modest: each fine-tune is a few GPU-hours, and

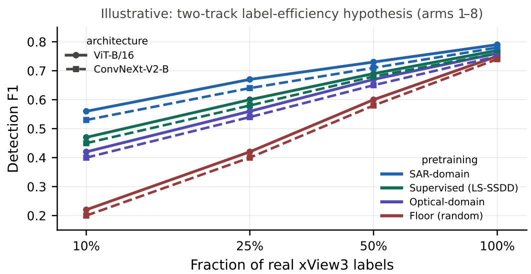

Figure 2. Illustrative two-track label-efficiency hypothesis (eight arms). Solid = ViT, dashed = ConvNeXt-V2; color = pretraining role. We hypothesize that SAR-domain pretraining beats optical for both architectures, with gaps largest at low label budgets.

the full eight-arm grid (both tracks, with seed reruns including the low-label cells) plus the two small LS-SSDD supervised backbones comes to roughly 293–360 GPU-hours, about three to four nights on the node—an order of magnitude below self-pretraining a SAR foundation model. This lower cost is a side benefit; as motivated in §1, we use downloaded checkpoints for comparability of pretraining recipes as published, not to save compute. The downloaded-backbone arms run first, since the main technical risk is the channel-format gap (whether each pretrained stem accepts the fixed [VH, VV, VH−VV] input); a quick load-and-train check de-risks the shared input per architecture before the full grid. We target the WACV 2027 Round-2 deadline (paper registration August 21, 2026; submission August 28, 2026, AoE).

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